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Task 4: Sentiment Analysis Web Application Using Flask and TextBlob

1. Introduction

Sentiment analysis is a natural language processing technique used to identify the emotional tone of a given text. It helps in classifying text into positive, negative, or neutral categories. In this task, a web application is developed using Flask that performs sentiment analysis on user-entered text using the TextBlob library.

2. Objective

The objectives of this task are:

- To develop a web application using Flask
 - To perform sentiment analysis on user-entered text
 - To classify the sentiment as positive, negative, or neutral
 - To display polarity and subjectivity scores of the text
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3. Tools and Technologies Used

- **Programming Language:** Python
 - **Framework:** Flask
 - **Libraries Used:**
 - TextBlob – for sentiment analysis
 - Flask – for web application development
 - **Frontend Technologies:** HTML and CSS
 - **IDE:** Visual Studio Code
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4. System Description

The system consists of a web interface where users can enter text. Upon submission, the backend processes the input text using TextBlob and analyzes its sentiment. The result is displayed on the same webpage along with polarity and subjectivity values.

5. Methodology

1. A Flask application was created using Python.

2. An HTML form was designed to accept user input.
 3. The entered text was processed using the TextBlob library.
 4. Polarity and subjectivity scores were calculated.
 5. Based on the polarity value, the sentiment was classified as positive, negative, or neutral.
 6. The results were displayed on the web page using HTML and CSS.
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6. Results and Observations

- The application successfully classified user-entered text into positive, negative, or neutral sentiment.
 - Polarity values ranged between -1 and +1, indicating the emotional tone of the text.
 - Subjectivity values indicated whether the text was opinion-based or factual.
 - The web interface was simple, user-friendly, and responsive.
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7. Conclusion

This task successfully demonstrated the development of a sentiment analysis web application using Flask and TextBlob. It helped in understanding web application development, text processing, and basic natural language processing concepts. The project enhanced practical knowledge of backend and frontend integration in Python.

8. Output

- Flask application (app.py)
- HTML template (index.html)
- CSS file (style.css)
- Web page displaying sentiment, polarity, and subjectivity